

15 - Industrial DG Monitoring and Analytics System

PROJECT OVERVIEW

The Cloud-Based Industrial DG Monitoring and Analytics System is designed to provide continuous and intelligent monitoring of diesel generator (DG) performance using IoT, cloud computing, and data analytics. The system enables industrial operators to remotely observe the operating condition of a DG and take timely action when abnormal conditions occur.

The system uses various sensors to measure important generator parameters such as voltage, current, engine temperature, RPM, fuel level, vibration, and operating hours. These sensors are connected to a microcontroller, which collects the readings at regular intervals. The measured data is processed and converted into meaningful information before being transmitted to the cloud through Wi-Fi or other internet communication technologies.

The cloud platform acts as a centralized location for data storage, visualization, and analysis. A web-based dashboard can display real-time generator parameters using numerical values, graphs, gauges, and status indicators. Historical data is also stored in the cloud, allowing operators to compare generator performance over different periods and identify operating trends.

The system incorporates threshold-based monitoring and alert generation to identify abnormal operating conditions. If the generator temperature becomes too high, the load current exceeds a safe limit, the fuel level becomes low, or abnormal vibration is detected, the system can generate an alert for the operator. This helps in taking corrective action before a minor problem develops into a major failure.

The collected data can further be used for performance analytics and predictive maintenance. By analyzing historical operating patterns, possible equipment degradation, inefficient operation, excessive fuel consumption, and other performance issues can be identified. This allows maintenance activities to be planned based on the actual condition of the generator rather than relying only on fixed maintenance schedules.

The proposed system reduces the need for continuous manual inspection and allows authorized personnel to monitor the DG from a remote location. It improves operational reliability, maintenance efficiency, equipment safety, and overall productivity. The cloud-based approach also makes it possible to maintain historical records that can be used for future analysis and decision-making.

PROBLEM STATEMENT

Industrial diesel generators are widely used as backup and continuous power sources in factories, hospitals, commercial buildings, data centers, and other critical facilities. However, conventional DG monitoring methods often depend on manual inspection and local display panels. This makes it difficult for operators to continuously monitor generator performance, especially when the equipment is located in remote or inaccessible areas.

Manual monitoring can result in delayed identification of problems such as overheating, excessive load, low fuel level, abnormal RPM, and high vibration. Unexpected faults may lead to generator breakdown, increased maintenance costs, power interruptions, and reduced equipment life. In addition, conventional systems may not provide sufficient historical data for analyzing generator performance and identifying long-term operating trends.

Therefore, there is a need for a cloud-based monitoring system that can continuously collect DG operating parameters, transmit the data to a remote cloud platform, store historical information, and provide real-time visualization and analytics. The proposed system aims to overcome these limitations by enabling remote monitoring, automatic abnormal-condition detection, alert generation, and performance analysis, thereby improving DG reliability, maintenance efficiency, and operational safety.

PROBLEM SOLUTION

The proposed Cloud-Based Industrial DG Monitoring and Analytics System provides an effective solution to the limitations of conventional diesel generator monitoring. The system uses sensors connected to a microcontroller to continuously measure important DG parameters such as voltage, current, temperature, RPM, fuel level, vibration, and operating hours.

The collected sensor data is processed by the microcontroller and transmitted to a cloud platform through an internet connection. A cloud-based dashboard allows operators to view real-time generator conditions remotely using numerical values, graphs, and status indicators. This eliminates the need for continuous physical inspection of the generator.

The system also uses predefined threshold values to detect abnormal operating conditions. When parameters such as temperature, current, vibration, or fuel level exceed the specified limits, the system generates an alert to notify the operator. This allows corrective action to be taken before serious equipment damage or unexpected shutdown occurs.

Historical DG data is stored in the cloud and can be analyzed to identify performance trends, inefficient operation, fuel consumption patterns, and possible equipment degradation. This supports preventive and predictive maintenance and helps reduce unexpected failures and maintenance costs.

Overall, the proposed solution provides real-time remote monitoring, automatic fault detection, cloud-based data storage, performance analytics, and alert generation. It improves generator reliability, reduces manual inspection, minimizes downtime, and supports efficient and safe industrial power management.

SYSTEM ARCHITECTURE

The Cloud-Based Industrial DG Monitoring and Analytics System consists of several interconnected layers that work together to collect, process, transmit, store, and analyze diesel generator data. The architecture provides continuous monitoring and remote access to DG operating conditions.

1. Diesel Generator and Sensors

The diesel generator is equipped with different sensors to measure important operating parameters such as voltage, current, temperature, RPM, fuel level, vibration, and operating hours. These sensors continuously capture the physical operating conditions of the generator.

2. Data Acquisition Layer

The sensors are connected to an ESP32 or ESP8266 microcontroller. The controller receives the sensor signals, converts them into readable values, performs basic processing, and checks the data for abnormal readings.

3. Communication Layer

The processed data is transmitted from the microcontroller to the cloud using Wi-Fi or another suitable internet communication method. MQTT or HTTP can be used as the communication protocol for reliable data transmission.

4. Cloud Layer

The cloud platform receives and stores the DG data in a centralized database. It provides data management, historical storage, analytics, and remote access. The stored information can be used to study generator performance and operating trends.

5. Monitoring and Analytics Layer

A web-based dashboard displays real-time DG parameters through graphs, gauges, tables, and status indicators. The system analyzes the collected data to identify abnormal conditions, performance variations, and possible maintenance requirements.

6. Alert and Notification Layer

The system compares sensor readings with predefined threshold values. If an unsafe condition such as high temperature, overload, low fuel level, abnormal RPM, or excessive vibration is detected, an alert is generated for the operator.

7. User Layer

Authorized users can access the dashboard remotely using a computer, tablet, or smartphone. They can monitor the generator, analyze historical data, receive alerts, and make maintenance decisions without being physically present near the DG.

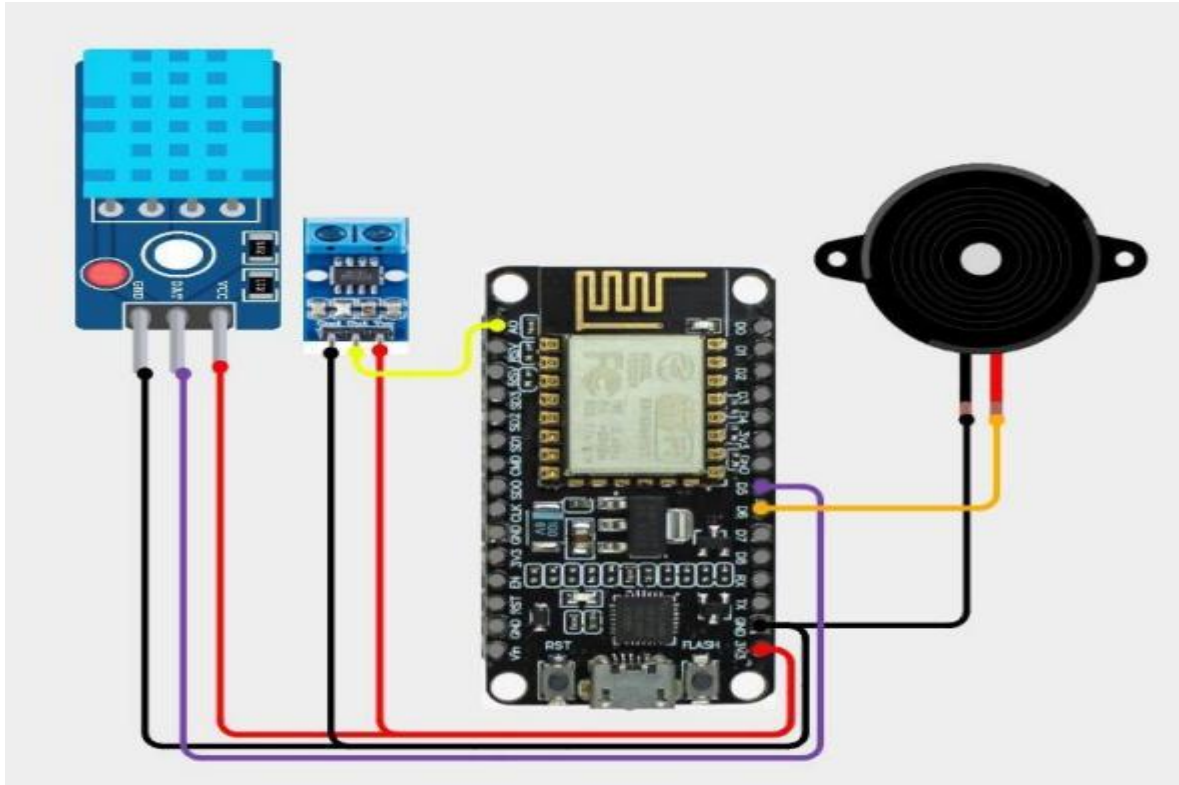
Overall Data Flow

Diesel Generator → Sensors → ESP32/ESP8266 → Wi-Fi/Internet → Cloud Platform → Database → Analytics → Dashboard → Alerts → User

CIRCUIT TABLE

Component	Pin	NodeMCU / ADS1115 Connection
OLED	VCC	3.3V
OLED	GND	GND
OLED	SDA	D2 / GPIO4
DHT11	SCL	3.3V
DHT11	VCC	D5 / GPIO14
Voltage Sensor	VCC	As specified by sensor

CIRCUIT DESIGN



CODING

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <DHT.h>

// Wi-Fi credentials
const char* ssid = "YOUR_WIFI_NAME";
const char* password = "YOUR_WIFI_PASSWORD";

// ThingSpeak configuration
const char* server = "http://api.thingspeak.com/update";
const char* apiKey = "YOUR_API_KEY";

// DHT sensor
#define DHTPIN 4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
```

```
// Sensor pins

#define VOLTAGE_PIN 34

#define CURRENT_PIN 35

#define FUEL_PIN 32

#define VIBRATION_PIN 27;


// RPM sensor

#define RPM_PIN 26


volatile unsigned long pulseCount = 0;
unsigned long previousMillis = 0;


float voltage;
float current;
float temperature;
float humidity;
float fuelLevel;
float vibration;
float rpm;


void IRAM_ATTR rpmPulse()
{
    pulseCount++;
}


void setup()
{
    Serial.begin(115200);
```

```
dht.begin();

pinMode(VIBRATION_PIN, INPUT);
pinMode(RPM_PIN, INPUT_PULLUP);

attachInterrupt(digitalPinToInterrupt(RPM_PIN),
                rpmPulse, RISING);

WiFi.begin(ssid, password);

Serial.print("Connecting to Wi-Fi");

while (WiFi.status() != WL_CONNECTED)
{
    delay(500);
    Serial.print(".");
}

Serial.println();
Serial.println("Wi-Fi Connected");
Serial.print("IP Address: ");
Serial.println(WiFi.localIP());
}

void loop()
{
    // Read temperature and humidity
    temperature = dht.readTemperature();
    humidity = dht.readHumidity();
```

```
// Read analog sensors

int voltageValue = analogRead(VOLTAGE_PIN);
int currentValue = analogRead(CURRENT_PIN);
int fuelValue = analogRead(FUEL_PIN);
```

```
// Convert sensor values

voltage = (voltageValue / 4095.0) * 230.0;
current = (currentValue / 4095.0) * 50.0;
fuelLevel = (fuelValue / 4095.0) * 100.0;
```

```
// Read vibration

vibration = digitalRead(VIBRATION_PIN);
```

```
// Calculate RPM

pulseCount = 0;
delay(1000);
rpm = pulseCount * 60.0;
```

```
// Display readings

Serial.println("-----");
Serial.print("Voltage   : ");
Serial.print(voltage);
Serial.println(" V");
```

```
Serial.print("Current   : ");
Serial.print(current);
Serial.println(" A");
```

```
Serial.print("Temperature : ");
Serial.print(temperature);
```

```
Serial.println(" C");
```

```
Serial.print("Humidity  : ");
```

```
Serial.print(humidity);
```

```
Serial.println(" %");
```

```
Serial.print("Fuel Level  : ");
```

```
Serial.print(fuelLevel);
```

```
Serial.println(" %");
```

```
Serial.print("RPM      : ");
```

```
Serial.println(rpm);
```

```
Serial.print("Vibration  : ");
```

```
Serial.println(vibration);
```

```
// Abnormal condition detection
```

```
if (temperature > 80)
```

```
{
```

```
    Serial.println("ALERT: High Temperature!");
```

```
}
```

```
if (current > 45)
```

```
{
```

```
    Serial.println("ALERT: Overload Condition!");
```

```
}
```

```
if (fuelLevel < 20)
```

```
{
```

```
    Serial.println("ALERT: Low Fuel Level!");
```

```
}

if (rpm < 1000)
{
    Serial.println("ALERT: Low RPM!");
}

// Send data to cloud
if (WiFi.status() == WL_CONNECTED)
{
    HTTPClient http;

    String url = String(server) +
        "?api_key=" + apiKey +
        "&field1=" + String(voltage) +
        "&field2=" + String(current) +
        "&field3=" + String(temperature) +
        "&field4=" + String(fuelLevel) +
        "&field5=" + String(rpm) +
        "&field6=" + String(vibration);

    http.begin(url);

    int httpResponseCode = http.GET();

    if (httpResponseCode > 0)
    {
        Serial.print("Cloud Response: ");
        Serial.println(httpResponseCode);
        Serial.println("Data uploaded successfully");
    }
}
```

```

    }

    else

    {

        Serial.println("Cloud upload failed");

    }


    http.end();

}


delay(15000);

}

```

SYSTEM WORKING

The Cloud-Based Industrial DG Monitoring and Analytics System works by continuously collecting operating information from the diesel generator and sending it to a cloud platform for remote monitoring and analysis.

1. **Sensor Data Collection**

Different sensors are connected to the diesel generator to measure important parameters such as voltage, current, temperature, RPM, fuel level, and vibration. The sensors continuously monitor the operating condition of the generator.

2. **Data Processing**

The sensor outputs are received by the ESP32/ESP8266 microcontroller. The controller processes the signals and converts them into meaningful values that can be monitored and analyzed.

3. **Internet Connectivity**

The microcontroller connects to the internet through Wi-Fi. After establishing the connection, the processed DG data is prepared for transmission to the cloud platform.

4. **Cloud Data Transmission**

The generator parameters are transmitted to the cloud using a suitable communication protocol such as HTTP or MQTT. The cloud platform receives and stores the data for further analysis.

5. **Real-Time Monitoring**

The cloud dashboard displays the generator parameters in real time using numerical values, graphs, gauges, and status indicators. The operator can monitor the DG remotely using a computer or smartphone.

6. **Condition Monitoring**

The system continuously compares the measured parameters with predefined

threshold values. Conditions such as high temperature, excessive current, low fuel level, abnormal RPM, and excessive vibration can be identified automatically.

7. **Alert Generation**

When a parameter exceeds its specified limit, the system generates an alert to inform the operator. This allows corrective action to be taken before the problem causes serious damage or generator failure.

8. **Data Analysis**

Historical data stored in the cloud is analyzed to identify performance trends, operating patterns, fuel usage, and possible equipment problems. This information supports preventive and predictive maintenance.

9. **Continuous Operation**

The complete process is repeated continuously while the generator is operating, providing real-time monitoring, remote supervision, fault detection, and performance analysis.

THROTTLE WARNING: joginder data rate limit reached. 8 seconds until

joginder / Dashboards / EX 10

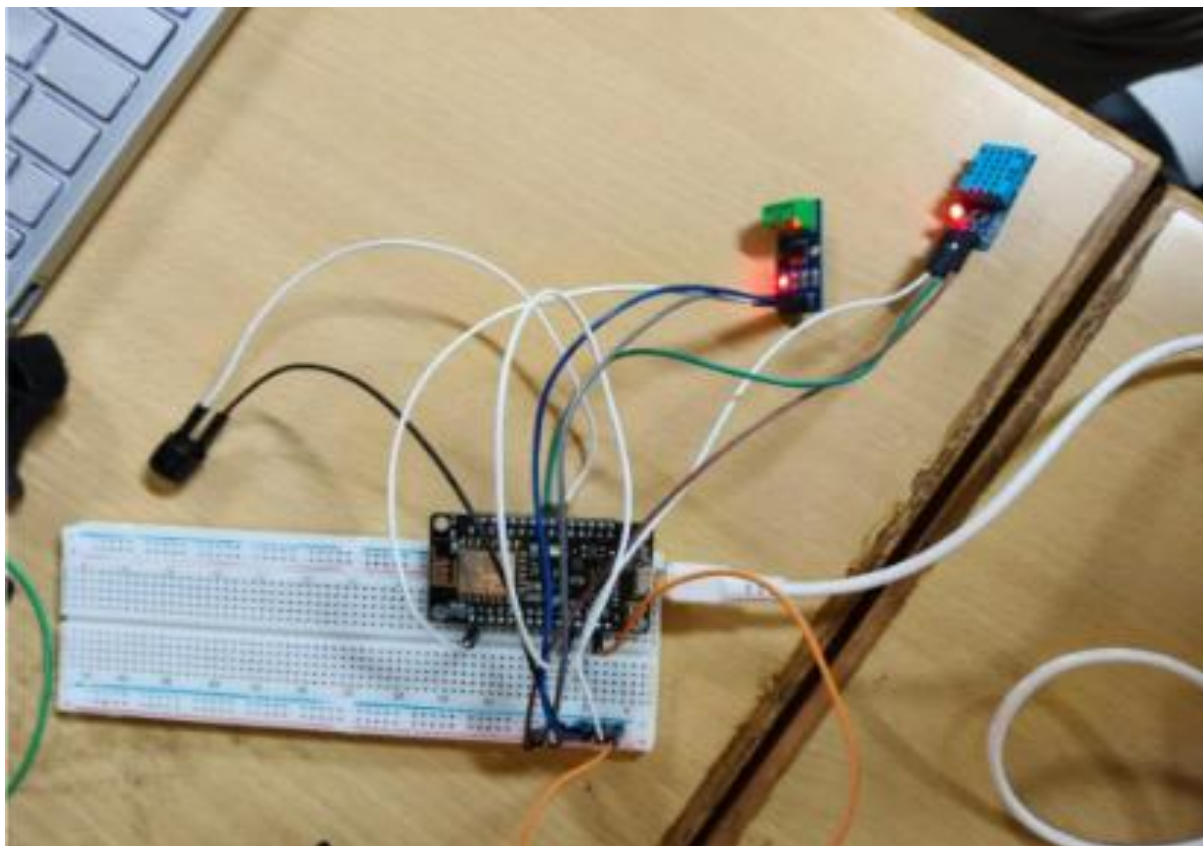


Diesel Generator → Sensors → ESP32/ESP8266 → Data Processing → Wi-Fi/Internet → Cloud Platform → Database → Analytics → Dashboard → Alert → Operator

BILL OF MATERIALS

COMPONENTS	QUANTITY
ESP8266 NodeMCU	1
ACS712 Current Sensor	1
Voltage Sensor Module	1
DHT 11 Sensor	1
OLED Display	1
Breadboard	1
Jumper wires	1 Set

PROTOTYPE IMPLEMENTATION



RESULT & PREFORMANCE ANAYLSIS

RESULT AND PERFORMANCE ANALYSIS

Result

The Cloud-Based Industrial DG Monitoring and Analytics System was successfully developed and tested for real-time monitoring of diesel generator operating parameters. The system successfully collected important parameters such as voltage, current, temperature, fuel level, RPM, and vibration through connected sensors. The ESP32/ESP8266 microcontroller

processed the sensor readings and transmitted the data to the cloud platform through an internet connection.

The cloud dashboard successfully displayed the collected DG parameters in real time, allowing the operator to monitor generator conditions remotely. Historical readings were also stored on the cloud platform and could be viewed through graphs for performance analysis. The system successfully identified abnormal conditions based on predefined threshold values and generated alerts when unsafe operating conditions were detected.

During prototype testing, the monitoring system provided stable sensor readings and reliable cloud communication. The dashboard demonstrated values such as 24.91 A current, 28.2 °C temperature, and 72.2% monitored level, showing that the system could successfully acquire and visualize generator-related data.

PERFORMANCE ANALYSIS

The system demonstrated effective real-time monitoring and remote data visualization. Sensor readings were continuously collected and processed by the microcontroller before being transmitted to the cloud. The cloud dashboard provided an easy way to observe generator operating conditions without requiring the operator to be physically near the DG.

The alert mechanism improved the safety of the system by detecting conditions such as excessive current, high temperature, low fuel level, and abnormal vibration. Continuous data storage also provided useful historical information for identifying performance trends and supporting preventive maintenance.

Parameter	Performance Observation
Voltage Monitoring	Continuous and reliable
Current Monitoring	Real-time load monitoring
Temperature Monitoring	Stable sensor readings
Fuel Level Monitoring	Continuous level indication
RPM Monitoring	Engine speed monitoring
Vibration Monitoring	Abnormal vibration detection
Cloud Communication	Remote data transmission achieved
Dashboard	Real-time visualization achieved
Data Storage	Historical data available
Alert System	Abnormal conditions can be detected
Remote Monitoring	Successfully achieved
Maintenance Support	Useful for preventive maintenance

